

CONTACT  
INFORMATION

## Adrian William Romero Jorge

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Date of Birth: 15.06.1998



EDUCATION

**Ph.D. Student**, Goethe University Frankfurt, Germany.  
*Supervisor: Prof. Dr. Elena Bratkovskaya, Prof. Dr. Laura Sagunski* **2023 - present**

**M.S. Physics**, University of Havana, Cuba.  
*Supervisors: Prof. Dr. A. Martínez Pérez, Prof. Dr. E. Rodríguez Querts* **2022 - 2023**

**B.S. Physics**, University of Havana, Cuba.  
*Supervisors: Prof. Dr. A. Martínez Pérez, Prof. Dr. E. Rodríguez Querts. Five-years course plus a thesis* **2016 - 2021**

POSTGRADUATES  
COURSES

**University of Havana & Higher Institute of Technologies and Applied Sciences, Cuba.**

- Introduction to General Relativity Theory
- Automation of Physics Experiments with the Arduino Platform
- Standard Model of Particles. Mathematical Methods
- Introduction to Quantum Fields in Classic Contexts
- Quantum Field Theory
- Cosmology
- Introduction to Standard Model of Particles
- Nonlinear Quantum Electrodynamics

COMPLEMENTARY  
EDUCATION

**Institute for Theoretical Physics, Goethe University, Frankfurt, Germany.**

- Physics of Strongly Interacting Matter
- Dynamical models for relativistic heavy-ion collisions

**Galileo Galilei Institute (GGI), Italy**

- School: Direct Detection across Dark Matter mass ranges 2025 (TME)
- School: Frontiers in Nuclear and Hadronic Physics 2026 (FNHP)

**Helmholtz Graduate School for Hadron and Ion Research (HGS-HIRE), Germany.**

- Making an Impact as an Effective Researcher
- Career and Leadership
- Leading Teams in a Research Environment
- Machine Learning from Scratch

PRESENTATIONS  
& EVENTS

**XXX Identification of Dark Matter 2026 (iDM)** **2025**  
*Talk: "Combined constraints on dark photons from high-energy collisions, cosmology, and astrophysics". Zaragoza, Spain.*

**Theory Meets Experiments 2025: Dark Matter Direct Detection (TME)**  
*Lightning Talk: "Search for dark photons in heavy-ion collisions ". Florence, Italy.* **2025**

**IV DarkMatter@Bonn Meeting (DMLab)**

Talk: “Combined constraints on dark photons from high-energy collisions, cosmology, and astrophysics”. Bonn, Germany. **2025**

**V Dark Matter Conference: From the Smallest to the Largest Scale (DM)**

Poster and Flash Talk: *Constraining Dark Photon Kinetic Mixing in Heavy-Ion Collisions from SIS to LHC energies*. Santander, Spain. **2025**

**XXXI International Conference on Ultra-relativistic Nucleus-Nucleus Collision (QM)**

Poster: *Electromagnetic emission from strongly interacting hadronic and partonic matter created in heavy-ion collisions*. Frankfurt, Hesse, Germany **2025**

**III Workshop on Heavy Ions Physics and Compact Stars (HIPSTARS)**

Talk: “Exploring Dark Photon Production and Kinetic Mixing Constraints in Heavy-Ion Collisions”. ICIMAF, Havana, Cuba. **2024**

**42<sup>nd</sup> International Conference on High Energy Physics (ICHEP)**

Talk: “Search for dark photon in heavy ion collisions”. Prague, Czech Republic. **2024**

**GET INvolved Program by GSI-FAIR**

Research training with Dr. Elena Brakovskaya. GSI, Darmstadt, Germany. **2022**

**II Workshop on Heavy Ions Physics and Compact Stars (HIPSTARS).**

Talk: “On the propagation of photons in rotating neutron stars including quantum and gravitational effects”. ICIMAF, Havana, Cuba. **2022**

**Proposals for the Award of the Nuclear Energy Agency and Advanced Technologies (AENTA).**

Talk: “Delay of the magnetized photon in astrophysical scenarios”. Cuban National Hotel, Havana, Cuba. **2021**

**I Workshop on Heavy Ions Physics and Compact Stars (HIPSTARS).**

Talk: “The photon time delay in strong magnetic field”. ICIMAF, Havana, Cuba. **2020**

**XV Symposium of the Cuban Physics Society.**

Poster: “The photon time delay in magnetized vacuum magnetosphere”. University of Havana, Havana, Cuba. **2020**

**VI International Sym. on Strong Electromagnetic Fields and Neutron Stars SMFNS). and V Caribbean Sym. on Cosmology, Gravitation, Nuclear and Astroparticle Physics (STARS).**

Talk: “The magnetized photon time delay”. Varadero, Cuba. **2019**

**SCHOOLS**

**III German-Cuban Summer School by DAAD-FAIR Program.**

Participation. ICIMAF, Havana, Cuba. **2023**

**European Summer School on Science Communication (ES3C) by DAAD-FAIR Program.**

Participation. FIAS, Frankfurt, Germany. **2022**

**II German-Cuban Summer School by DAAD-FAIR Program.**

Talk: “Photon time delay in neutron stars magnetosphere”. ICIMAF, Havana, Cuba. **2022**

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|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
|                                          | <b>I German-Cuban Summer School by DAAD-FAIR Program.</b><br><i>Participation. Havana, Cuba.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2019 |
| WORKSHOPS AS<br>UNDERGRADUATE<br>STUDENT | <b>Student Scientific Forum of Havana University.</b><br><i>Participation. University of Havana, Havana, Cuba.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2022 |
|                                          | <b>XXXVII Student Scientific Workshop – Physics Faculty.</b><br><i>Talk: “Photon time delay in neutron stars magnetosphere”.</i><br><i>Faculty of Physics, University of Havana, Havana, Cuba.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2021 |
|                                          | <b>VI Student Congress of Scientific and Technological Research (CEICYT).</b><br><i>Online Talk: “Photon time delay modeling in the magnetosphere of pulsars with ultra-intense magnetic fields”. Dominican Republic.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2021 |
|                                          | <b>XXXVI Student Scientific Workshop – Physics Faculty.</b><br><i>Participation. Faculty of Physics, University of Havana, Havana, Cuba.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2020 |
|                                          | <b>XXXV Student Scientific Workshop – Physics Faculty.</b><br><i>Talk: “Delay of magnetized photons”.</i><br><i>Faculty of Physics, University of Havana, Havana, Cuba.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2019 |
|                                          | <b>XXXIV Student Scientific Workshop – ICIMAF.</b><br><i>Talk: “Delay of photons in magnetized vacuum”.</i><br><i>Faculty of Physics, University of Havana, Havana, Cuba.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2018 |
|                                          | <b>XXXIII Student Scientific Workshop – Physics Faculty.</b><br><i>Talk: “Quantum dynamics of vibrations between the rings of benzene”.</i><br><i>Faculty of Physics, University of Havana, Havana, Cuba.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2017 |
| PEER-REVIEWED<br>PUBLICATIONS            | <b>A. W. Romero Jorge</b> , L. Sagunski, Guan-Wen Yuan, T. Song and E. Brakovskaya, “ <i>Combined constraints on dark photons from high-energy collisions, cosmology, and astrophysics</i> ”, Phys. Rev. D <b>113</b> , 113, 055052, arXiv.2601.15066, 2026.<br><br><b>A. W. Romero Jorge</b> , E. Brakovskaya, T. Song and L. Sagunski, “ <i>Constraints on Kinetic Mixing of Dark Photons from Dilepton Spectra</i> ”, Phys. Rev. C, <b>112</b> , 054905, arXiv.2507.11163, 2025.<br><br><b>A. W. Romero Jorge</b> , T. Song, Q. Zhou and E. Brakovskaya, “ <i>Electromagnetic emission from strongly interacting hadronic and partonic matter created in heavy-ion collisions</i> ”, Phys. Rev. C, <b>111</b> , 064904, arXiv.2503.05253, 2025.<br><br><b>A. W. Romero Jorge</b> , A. Pérez Martínez, and E. Rodríguez Querts, “ <i>Nonlinear electrodynamics for the vacuum of Dirac materials. Photon magnetic properties and radiation pressures</i> ”, Phys. Rev. A, <b>110</b> , 012201, arXiv.2402.01042 , 2024. |      |
| CONFERENCE<br>PROCEEDINGS                | <b>A. W. Romero Jorge</b> , E. Brakovskaya, T. Song, and L. Sagunski, “ <i>Exploring Dark Photon Production and Kinetic Mixing Constraints in Heavy-Ion Collisions</i> ”, Astron. Nachr., <b>346</b> : e20240132, arXiv.2412.02536, 2024.<br><br><b>A. W. Romero Jorge</b> , E. Brakovskaya, and L. Sagunski, “ <i>Search for dark photons in heavy-ion collisions</i> ”, Proceedings of 42nd International Conference on High Energy Physics, PoS (ICHEP2024), <b>290</b> , arXiv.2409.20141, 2024.<br><br>A. Pérez Martínez, M. Pérez García, E. Rodríguez Querts and <b>A. W. Romero Jorge</b> , “ <i>Vacuum</i>                                                                                                                                                                                                                                                                                                                                                                                                       |      |

*properties and astrophysical implications”, 16th Marcel Grossmann Meeting, 3756-3761, 2023.*

**A. W. Romero Jorge**, E. Rodríguez Querts, A. Pérez Martínez and H. Pérez Rojas, “*The photon time delay in magnetized medium magnetosphere*”, Astron. Nachr., **342**, 382-387, 2021.

**A. W. Romero Jorge**, E. Rodríguez Querts, A. Pérez Martínez et al., “*The photon time delay in magnetized vacuum magnetosphere*”, Astron. Nachr., **340**, 852-856, arXiv.1912.02904 2020.

HONORS  
& AWARDS

**Giersch Excellence Award 2025.**

*The grant is awarded for outstanding work in the thesis project in the past year.* **2024, 2025**

**Karin and Carlo Giersch Scholarship.**

*Awarded for Ph.D. studies in physics* **2023**

**Graduated with Honors (summa cum laude).**

*Awards for the best graduate of the career and most outstanding results in the scientific research. Faculty of Physics, University of Havana.* **2021**

**Award for the scientific result of the science and technological innovation system of the Nuclear Energy and Advance Technologies Agency (AENTA).** **2021**

*To the work “Magnetized photon delay in astrophysical scenarios” by the authors A. W. Romero Jorge, E. Rodríguez Querts and A. Pérez Martínez.*

**First place in the Student Scientific Workshop – Physics Faculty.**

*At the theoretical physics committee.* **2021**

TEACHING  
EXPERIENCE

**Faculty of Physics, University of Havana, Havana, Cuba.**

*Teaching Assistant* **2018-2019**

Experimental physics laboratory (Mechanics) and on the computational methods course (C++).

**High School Institute José Martí, Havana, Cuba.**

*Part time teacher* **2019**

Teaching Classical Mechanics to 10th grade.

PROFICIENCY

|                              |   |                                                                                                                   |
|------------------------------|---|-------------------------------------------------------------------------------------------------------------------|
| <b>Languages</b>             | : | Spanish (native), English (oral,writing: fluent),<br>German (A1), Japanese (begginer)                             |
| <b>Programming Languages</b> | : | Fortran, C++, Python, Java, XML, Arduino, Wolfram                                                                 |
| <b>Softwares/Tools</b>       | : | L <sup>A</sup> T <sub>E</sub> X, Origin, Wolfram Mathematica, Maple, Android Studio,                              |
| <b>Hard Skills</b>           | : | Drawing, Video Editing, Digital Design, Web & App development                                                     |
| <b>Soft Skills</b>           | : | Teamwork, Adaptability, Time management, Creativity,<br>Decision-making, Analytical thinking, Attention to detail |